IEEE JOURNAL OF BIOMEDICAL AND HEALTH INFORMATICS, VOL. 28, NO. 6, JUNE 2024

A Two-Stage Differential Privacy Scheme for Federated Learning Based on Edge Intelligence

Li Zhang, Jianbo Xu, Audithan Sivaraman, *Member, IEEE*, Jegatha Deborah Lazarus, *Member, IEEE*, Pradip Kumar Sharma, *Senior Member, IEEE*, and Vijayakumar Pandi, *Senior Member, IEEE*

***Abstract*—The issue of data privacy protection must be considered in distributed federated learning (FL) so as to ensure that sensitive information is not leaked. In this article, we propose a two-stage differential privacy (DP) framework for FL based on edge intelligence. Various levels of privacy preservation can be provided according to the degree of data sensitivity. In the first stage, the randomized response mechanism is used to perturb the original feature data by the user terminal for data desensitization, and the user can self-regulate the level of privacy preservation. In the second stage, noise is added to the local models by the edge server to further guarantee the privacy of the models. Finally, the model updates are aggregated in the cloud. In order to evaluate the performance of the proposed end-edge-cloud FL framework in terms of training accuracy and convergence, extensive experiments are conducted on a real electrocardiogram (ECG) signal dataset. Bi-directional long-short-term memory (BiLSTM) neural network is adopted to training classification model. The effect of different combinations of feature perturbation and noise addition on the model accuracy is analyzed depending on different privacy budgets and parameters. The experimental results demonstrate that the proposed privacy-preserving framework provides good accuracy and convergence while ensuring privacy.**

***Index Terms*—Differential privacy, edge computing, federated learning, smart healthcare.**

Manuscript received 30 March 2023; revised 7 July 2023; accepted 12 August 2023. Date of publication 18 August 2023; date of current version 6 June 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 61872138, in part by the Natural Science Foundation of Hunan Province under Grant 2021JJ30278, and in part by the Scientific Research Fund of Hunan Provincial Education Department under Grant 22B0497. *(Corresponding author: Jianbo Xu.)*

Li Zhang and Jianbo Xu are with the School of Computer Science and Engineering, Hunan University of Science and Technology, Xiangtan 411201, China, and also with the Hunan Key Laboratory for Service Computing and Novel Software Technology, Xiangtan 411201, China (e-mail: lzhang@mail.hnust.edu.cn; jbxu@hnust.edu.cn).

Audithan Sivaraman is with the Netaji Subhas University of Technology, New Delhi 110078, India (e-mail: audithan.sivaraman@nsut.ac.in).

Jegatha Deborah Lazarus and Vijayakumar Pandi are with the Department of Computer Science and Engineering, University College of Engineering Tindivanam, Tindivanam 604001, India (e-mail: blessedjeny@gmail.com; vijibond2000@gmail.com).

Pradip Kumar Sharma is with the Department of Computing Science, University of Aberdeen, AB24 3UE Aberdeen, U.K. (e-mail: pradip.sharma@abdn.ac.uk).

Digital Object Identifier 10.1109/JBHI.2023.3306425

I. Introduction

At present, artificial intelligence (AI) and sensor network are being widely used in the medical industry, which makes healthcare services truly intelligent. By using the Internet of Things technology and healthcare platform including massive health archives, the interaction and information sharing between patients and medical-related entities such as medical staff, medical institutions, and medical equipment can be realized. By machine learning from a large number of health samples, more accurate auxiliary diagnosis and treatment can be achieved. Currently, AI has a series of successful applications in medical diagnosis, such as skin cancer detection, chronic disease identification, pathological analysis, rare disease analysis [1], [2], etc.

The problem of data islands in healthcare field is an intractable problem, the emerging FL can solve the problem very well. FL is a framework in which a central server is responsible for training a high-quality and sharing-enabled global model from massive samples scattered across different clients [3], [4]. FL is a feasible approach to organize multiple medical institutions to jointly train models without disclosing any original healthcare data. Currently, there are already various applications in health industry [5], [6], [7]. However, FL also faces three major challenges: 1) Statistical feature is incomplete in an individual database of each patient. The data distribution of all clients varies greatly, and any available personal dataset is far from being representative samples of the overall distribution. 2) Communication efficiency is probably low. It requires effective coordination among a large number of clients in the federated training process, which causes frequent communication. 3) Privacy and security are still not ignored. It is impossible to ensure that all clients are absolutely reliable. How to prevent malicious clients from collecting model parameters for inference attacks or interfering the training process is also a problem to be addressed in FL [8], [9], [10], [11].

From the perspective of network system architecture, edge computing and edge intelligence have been applied in several industries [12], [13]. As shown in Fig. 1, it is a diagram of a collaborative end-edge-cloud smart healthcare architecture. By using wearable devices, patients can collect physiological signals such as blood sugar, blood pressure, blood oxygen saturation, and electrocardiogram, etc., which can then be transmitted to the hospitals or health institutions by the way of mobile devices, the gateways and base stations. In the end-edge-cloud

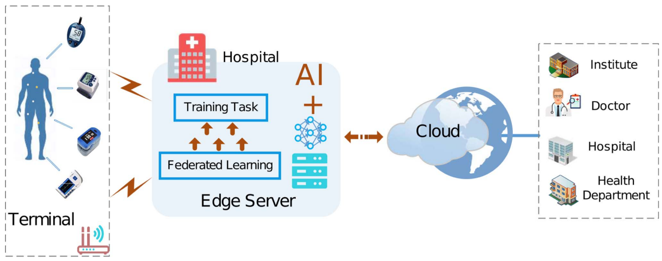


Fig. 1. Smart healthcare architecture based on end-edge-cloud collaboration.

architecture, health institutions can act as the edge servers for local model training, meanwhile, be considered as clients of cloud servers. After completing the collaborative training between health institutions and cloud, the final global models can be released on the relevant platforms for downloading and sharing by authorized academic institutions, health institutions and health management administrations.

As a vital physiological indicator of the human body, ECG signal is quite important. ECG monitoring has become an essential means for clinical diagnosis and health monitoring. Due to the critical property of ECG QRS wave, the automatic detection of QRS wave has attracted much attention [2], [14]. The occurrence time of QRS complex waves and their characteristic shapes provide the judgment basis for ECG monitoring and diagnostic systems. However, the eigenvalue extracted from the ECG signal contains the user's private information. For example, Huang et al. proposed to adopt ECG signal for identity authentication [15]. Thus, it can be seen that ECG signal is related to the important personal information. Protection of sensitive information is indispensable during the ECG signal processing. In this article, ECG signal detection is selected as our experimental scenario.

In order to prevent malicious clients from inferring other clients' data from models, there are two commonly used theoretical mechanisms, i.e. DP and multi-party computing (SMC). In DP mechanism, although disturbance including randomized response (RR) or noise addition are imposed on data, an undistinguishable model can still be trained whatever the data is original or not. Thereby, DP can be used for privacy-preserving computation [16], [17]. In FL based on DP, because noises addition brings negative affect on the speed of model convergence, then it may require more training epochs to achieve ideal accuracy [16]. The main problem and technical challenge of FL based on DP is how to enhance the privacy-preserving level of sensitive information without the loss of performance on accuracy and convergence.

In this article, we propose a two-stage DP for FL that can be suitable for the end-edge-cloud collaborative smart healthcare system as shown in Fig. 1. Different perturbance approaches are introduced in two stages. In the first stage, the randomized response technology is used at the user terminal for perturbing and obfuscate the most sensitive data features so as to provide autonomous and controllable privacy guarantees before sensitive information is sent to the edge server. In the second stage, a common classification model is trained by collaborating multiple edge servers deployed in different hospitals and cloud server. Before uploading the local model, the edge servers add noise to the model to improve the model security. Depending on different application scenarios, the first or second stage privacy protection scheme can be can be adopted separately or combined freely, which makes it more flexible to adjust privacy budgets and protection levels.

II. Related Work

With the diversification of model attacks, protection of private data is far from being as simple as hiding sensitive attributes directly in the data. Because the model is the nonlinear mapping of the original data, so adversaries can infer the original data from the model. The DP proposed by Dwork et al. solves this problem [16]. DP comes from the concept of semantic security in cryptography, where an attacker cannot distinguish the encryption results of different plaintexts. In DP mechanism, it is required that an attacker cannot extrapolate corresponding relationship between results and datasets according to the published results. The key idea is that the released results do not vary significantly due to the addition of the random noises, whatever an individual record is in the dataset or not. Furthermore, a quantitative model for the degree of privacy leakage is defined in this mechanism. Because the change of an individual record does not distinctively affect the data query results, an attacker cannot infer the privacy information of a single sample with a significant advantage, so the DP model does not depend on the background knowledge possessed by the attacker and provides a higher level of semantic security for privacy information. Thus, it has been widely used as an innovation for privacy protection.

Local DP (LDP) is an effective privacy-preserving method without a trustworthy the-third-party while data availability is still guaranteed. It also uses strictly mathematically definition [17]. Several companies such as Google, Apple and Microsoft have applied LDP mechanisms in their products for data statistics and model training. Some LDP algorithms have been proposed in academia, where the RAPPOR algorithm is a classical solution for privacy-preserving frequency statistics, which has small error and high data availability, but it requires prior prediction of candidate attribute values [19]. As mentioned earlier, federal learning can also be combined with DP during training process. For instance, model parameters generated by CNN network is usually superimposed with Gaussian or Laplacian noise for privacy security [18].

By adding noises to data or using generalization methods to mask certain sensitive attributes, the incredible third-parties cannot distinguish and restore data, which achieve privacy preservation. Huang et al. [20], Xia et al. [21] and Shin et al. [22] implemented DP-based security by means of input perturbation for simple training tasks such as logistic regression, K-means clustering, and matrix decomposition. However, perturbation to data brings negative impact on model convergence. Therefore, there exists some challenges in LDP-based federated leaning. Firstly, a much larger amounts of samples are required for obtaining the equivalent accuracy compared to the noiseless model trainings. Secondly, a balance between usability and privacy is more difficult to achieve for high-dimensional models. In contrast to LDP, global DP (GDP) allows a trustworthy the-third-party to collect the raw data and perturb the aggregated datasets or the models.

Abadi et al. designed a perturbation method for model training. An accumulator is used to calculate the sum of privacy budget after each epoch. Once the budget is exceeded, the use of samples will be stopped immediately, so the control of DP is more accurate [18]. Geyer et al. proposed to collect model updates and hide the contribution of each node by adding noise one by one by the server. This centralized noise addition scheme can achieve user-level DP, which means any participants including the previous ones in the training are not exposed [23]. Hu et al. introduced DP mechanism into the FL of medical and health data, so as to solve some issues such as the security of sensitive medical data, resource constraints when data in multiple sites is transmit and integrated, the failure of single point etc. [24]. Sun et al. proposed an improved FL framework with LDP, which adaptively perturbs the weight of the deep neural network model while the weight coefficient remains within a certain range. Then, parameter shuffling is used for model aggregation, which enhanced privacy preservation. Ultimately, a high accuracy model is acquired [25]. Wei et al. added noise to the model and adjusted the noise parameters to satisfy DP. The relationships among the model convergence performance, privacy protection level, the number of participants and the maximum aggregation epoch in federated training are furtherly analyzed [26].

III. Two-Stage Federated Learning Based on Differential Privacy

A. System Framework and Security Requirements

In the scenario of smart healthcare, the existing FL frameworks based on the edge-cloud collaboration is deficient in considering privacy for terminal patients. Healthcare institutions acting as edge servers usually need to collect raw data from patients for FL. The healthcare institutions are usually assumed to be credible absolutely in the edge-cloud architecture. However, the communication environment becomes exceptionally complex with the popularity of wireless body area networks (WBAN), which brings tremendous threats for the data privacy of terminal patients. Since samples coming from a certain terminal patient are limited and cannot involve most of the features of a disease, thus, it is not feasible to perform FL by terminal patients directly. Furthermore, healthcare institutions are not absolutely trustworthy in a real-world environment, it is necessary to provide terminal patients with heterogeneous privacy protection according to the data sensitivities and trustworthiness of institutions, which facilitate dynamic adjustability and adaptivity of privacy protection intensity.

To achieve heterogeneous privacy preservation for terminal patients, an end-edge-cloud FL framework based on DP is proposed. The framework consists of three types of entities: 1) *Terminal user:* as the owners of the original data; 2) *Edge server:* trains the local model on their medical datasets; 3) *Cloud server:* acts as a system initiator and a model aggregator.

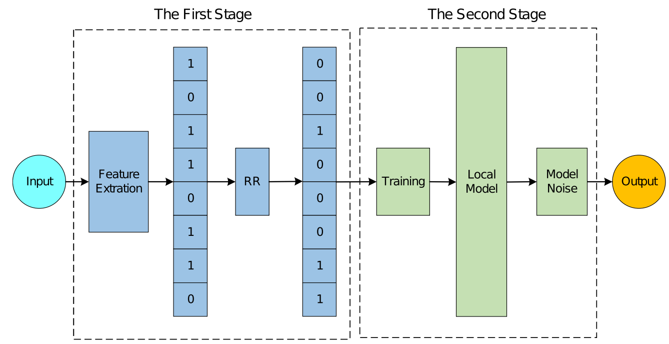


Fig. 2. Flow of the two-stage FL based on DP.

Throughout the FL training process, it is assumed that there probably exists external adversaries only interested in the privacy of users' data. Both of edge servers and cloud server are honest but curious, and they attempt to snoop on the privacy of terminal users although they can execute the training process correctly. For terminal users, they are only responsible for providing data sources for FL, they needn't to participate in the subsequent training process, nor need to interact with other users, so it is considered that they do not threat the data security. Therefore, the terminal users are assumed to be honest and trustworthy. Nevertheless, misbehaviors and deliberate disruption of model training such as poisoning attacks and tampering attacks by external adversaries or participants in FL are out scope of the proposed scheme.

B. Overview of the Proposed Scheme

The idea of privacy preservation in the end-edge-cloud framework is inspired by the concept of LDP and GDP. Under the premise of no any trustworthy servers, it is necessary for users to perform some perturbations on the original data before uploading them to the edge-servers. Because the local models trained by the edge servers contain certain knowledge properties that are associated with terminal users' data, noise is further superimposed on the local model to enhance privacy. The overall flow of the algorithm is shown in Fig. 2.

1) In the first stage, the terminal users extract the data features and transform them to binary vectors firstly, then perturbs the bits of the eigenvectors by randomized response algorithm, ultimately, submits the perturbed feature data to the edge server instead of submitting the original data. In this stage, the level of privacy preservation depends on bits perturbation intensity, which is controlled by the terminal users.
2) In the second stage, the cloud server coordinates multiple edge servers to train a common classifier on their local datasets. In order to enhance the privacy of the eigenvectors, a noise addition method is introduced to perturb the local model parameters in this stage.

C. DP-Based Algorithm for the First Stage

In this article, a DP mechanism based on randomized response technique is used in the first stage. It mainly relies on the post-processing invariance and combinatoriality of DP to ensure that the perturbant output of the first stage will not change the DP effect when they are applied to the convolutional neural network in the second stage.

For one-dimensional time-series data (e.g., ECG signals), we firstly pre-process the original signals to extract statistical eigenvectors such as time-frequency features, and then randomize the response to the feature data. The main steps of randomized perturbation are described as follows.

1) *Step 1:* Normalization of the feature data. In our algorithm, the terminal user firstly performs feature extraction operation on the raw signal and transform these features into a one-dimensional vector. Before the binary transformation, the eigenvalues are firstly normalized by using the z-score algorithm to ensure that all of them are within a certain range.
2) *Step 2:* Transformation of eigenvalues into binary form. The length of the binary bits depends on the z-score normalized range. Since the normalized eigenvalue is still floating point, the binary string needs to consist of a sign bit (1 for negative, 0 for positive), an integer part, and a fractional part. The bit length of the fractional part determines the data precision.
3) *Step 3:* Concatenation of binary strings. All of binary strings are concatenated into a long string so as to avoid the loss of privacy due to the combinatorial properties of DP mechanism. Given the privacy budget ε and the probability p of randomized response calculated according to ε. It is assumed that the bit length of a single eigenvalue after binarization is n and the length of eigenvector is c, then the total length of the concatenated binary string is cn. If each eigenvalue is randomized individually, the privacy budgets of all randomization operations will be superimposed, and the final privacy loss will be $c \times \varepsilon$.
4) *Step 4:* Selection and optimization of randomization probability. For two adjacent eigenvectors, after a series of operations, i.e. normalization, binarization and concatenation, two sets of different outputs are generated. Assuming that the two sets of outputs differ by at most Δf bits, the sensitivity is Δf. In the proposed scheme, two adjacent concatenated binary strings can differ by at most cn bits, so the sensitivity $\Delta f = cn$. In the randomized response algorithm, the randomization probability p usually represents the probability of responding the true value of the original bit, q is the probability of responding the false value. Let s_i represent an instance of eigenvectors, X is the binary string corresponding to s_i, and its length is cn. $X[i]$ represents the i-th bit, $X'[i]$ represents the i-th bit after perturbation. It is supposed that the sensitivity of X is $\Delta f = cn$. If the bits of X are perturbed according to (1). It can be proved that the perturbation mechanism satisfies the requirement of ε-DP when $p = \frac{e^{\varepsilon/cn}}{1+e^{\varepsilon/cn}}$, $q = \frac{1}{1+e^{\varepsilon/cn}}$. The specific proof is presented in Appendix A.

$$\Pr\left[X'[i] = 1\right] = \begin{cases} p, & \text{if } X[i] = 1 \\ q, & \text{if } X[i] = 0 \end{cases} \tag{1}$$

$$\varepsilon = \ln\left(\frac{p(1-q)}{(1-p)q}\right) \tag{2}$$

In the above disturbance mechanism, the probability of maintaining the original bit 1 and 0 are p and $1-q$ respectively (here, $1-q=p$), while the probability of responding bit 1 to 0 and bit 0 to 1 are $1-p$ and q respectively (here, $1-p=q$). Therefore, it can be seen that the probability of a true response is p and the probability of a false response is q whether the bit 0 or bit 1 is perturbed. However, it has been found that the number of bits 0 is often more than the number of bits 1 when the concatenated binary string is very long. At this point, perturbance to bit 0 or bit 1 with the same probability may lead to a reduction in the data availability. Therefore, we can learn from the idea of Optimized Unary Encoding (OUE) [27] to perturb 0 and 1 with different probabilities, so as to maintain the original state of bit 0 as much as possible by adjusting the privacy budget, i.e., decreasing the perturbation probability from 0 to 1, while keeping the perturbation probability from 1 to 0 constant. Assuming that the sensitivity is $\Delta f = cn$, according to (1), when the probability is set as $p = \frac{1}{2}$, $q = \frac{1}{1+e^{2\varepsilon/cn}}$, it can be proved that the perturbation mechanism satisfies ε-DP. The proof process is similar with that in Appendix A, so it is skipped here.

We introduce a new privacy budget coefficient α for probabilistic selection. α provides more flexibility in choosing the randomization probabilities. By increasing α, we can increase the probability of a true response to bit 0, introducing Theorem 1 as follows.

Theorem 1 (Random response algorithm after coefficientization): For any input s_1 and s_2, when

$$\Pr[X[s_1]=1|s_1] = \frac{1}{1+\alpha}, \Pr[X[s_1]=1|s_2] = \frac{1}{1+\alpha e^{\frac{2\varepsilon}{cn}}},$$

$$\Pr[X[s_2]=0|s_1] = \frac{\alpha e^{\frac{2\varepsilon}{cn}}}{1+\alpha e^{\frac{2\varepsilon}{cn}}}, \Pr[X[s_2]=0|s_2] = \frac{\alpha}{1+\alpha},$$

the algorithm satisfies ε-DP, the specific proof is given in Appendix B.

Theorem 1 is suitable for the original binary data which contain more numbers of bits 0 than bits 1. In proposed algorithm, introduction of the coefficient α makes the perturbation also applicable to high-sensitivity situations. It is not necessary to set an excessively high privacy budget ε because of too high sensitivity. The disturbance probability can be adjusted by changing the coefficient α instead of privacy budget ε. The main idea of the proposed algorithm is that retaining the original state of bits 0 as much as possible, meanwhile, tolerating the loss of data worth caused by the increase in the probability of bit 1 being responded to bit 0.

TABLE I
NOTATIONS

Symbol	Description
$\mathcal{P}$	the set of all edge servers
$\mathcal{P}_t$	the set of edge servers participating in the t-th round
N	the total number of all edge servers
S	the number of edge servers selected by cloud server
T	the total number of global rounds
T_i	the number of global rounds the i-th edge server participates
$\mathcal{L}(\cdot)$	loss function
t	the index of the t-th global round
d	the total size of data samples owned by all edge servers
d_i	the size of data samples owned by the i-th edge servers
$\hat{\mathbf{w}}$	the global model parameters
$\mathbf{w}_i$	the local model parameters of the i-th edge servers
$\mathbf{w}^*$	the optimal global model parameters

Algorithm 1: DP-based FedAvg with Selected Clients.

1: Cloud server Initialize and distribute $\hat{\mathbf{w}}^0$, T, ρ, ε, δ to all edge servers
2: **for** $t = 0$,1 to T **do**
3: **Cloud server:**
4: Select S online edge servers as a group $\mathcal{P}_t \subset \mathcal{P}$
5: **for** each edge server $i \subset \mathcal{P}_t$ in parallel **do**
6: $\hat{\mathbf{w}}_i^{t+1} \leftarrow \mathbf{clientUpdate}(i, \hat{\mathbf{w}}^t)$
7: **end for**
8: $\hat{\mathbf{w}}^{t+1} \leftarrow \frac{N}{S} \sum_{i \in \mathcal{P}_t} \frac{d_i}{d} \hat{\mathbf{w}}_i^{t+1}$
9: distribute $\hat{\mathbf{w}}^{t+1}$
10: **Edge server:**
11: **clientUpdate**(i, $\hat{\mathbf{w}}^t$)
12: $\mathbf{w}_i^{t+1} \leftarrow \arg\min_{\mathbf{w}_i} (\mathcal{L}_i(\mathbf{w}_i) + \frac{\rho}{2} \|\mathbf{w}_i - \hat{\mathbf{w}}^t\|^2)$
13: $\hat{\mathbf{w}}_i^{t+1} \leftarrow \mathbf{w}_i^{t+1} + \mathbf{n}_i^{t+1}$
14: **return** $\hat{\mathbf{w}}_i^{t+1}$
15: **end for**

D. DP-Based Algorithm for the Second Stage

Before describing it in detail, some useful notations are defined as shown in Table I.

1) Description of DP-Based FL Algorithm: The process of DP-based FL is as Algorithm 1. Before training, the cloud server is responsible for generating some public parameters, such as the initialized global model $\hat{\mathbf{w}}^0$, preset global rounds T, regularization coefficients ρ, DP budget ε and slack δ. Then these parameters are distributed to all edge servers. In the proposed algorithm, it needn't all of edge servers to participate in each global round, the cloud server only selects S edge servers for model aggregation. In FL, due to insufficient computing power or network delay, the cloud server often needs to wait for some local model updates for a long time, therefore, the selection mechanism can effectively avoid this inefficient model aggregation to some extent. How to select the participants for each global round, there are several strategies such as round-robin mechanism or choosing the first S model updates that arrived at the cloud server. The specific strategy for selection is not the focus of the proposed algorithm.

For ease of analysis, simple round-robin mechanism is adopted. First, the cloud server broadcasts the initial model and related parameters. After receiving these parameters, the edge servers begin to train local models on their own datasets, and then add Gaussian noise on models according to Theorem 2 before uploading them to the cloud server. After receiving the local models of S edge servers, the cloud server uses the traditional weighted average method based on the amount of data for aggregation. In order to ensure that the strategy of participants selection is an unbiased estimation of the weighted average strategy of all edge servers, the aggregated model is multiplied by a coefficient N/S which is the reciprocal of participant sampling rate in selection strategy. The detail description can be referred in the proof of Lemma 1. After the cloud server completes the aggregation, the global model is broadcasted to all edge servers, then the next iteration continues.

Lemma 1 (Unbiased estimation of the client selection policy): The weighted average strategy based on the round-robin mechanism in Algorithm 1 is estimated unbiasedly with the weighted average strategy base on all edge servers.

$$\mathrm{E}\{\bar{\mathbf{w}}_{i \in \mathcal{P}_t}^t\} = \mathrm{E}\{\bar{\mathbf{w}}_{i \in \mathcal{P}}^t\} \tag{3}$$

The specific proof can be seen in Appendix C.

2) DP Mechanism: Currently, there mainly exist two different mechanisms for DP-based FL, i.e., adding Laplace noise or Gaussian noise to the model or gradient. The Gaussian mechanism is adopted in the proposed scheme in view of its better performance in convergence of model, the servers perform T rounds of global aggregation, and S edge servers are selected for each aggregation, then the sampling rate of edge servers in each round is S/N, and the total number of times each edge server i participates in global aggregation is $T_i = ST/N$ on average. The specific Gaussian mechanism is defined as follows.

Theorem 2 (Gaussian mechanism [18]): There exist two constants c_1 and c_2 such that given sampling rate q of the samples in each edge server, the total rounds T_i edge server i participates in the global aggregation, and the condition $\varphi \geq 1$, for any $\varepsilon < c_1 q^2 T_i$ and $\delta > 0$, if the Gaussian noise mechanism is set as $\mathcal{M} = \mathcal{N}(0, \sigma^2 \varphi^2)$, Algorithm 1 satisfies (ε, δ)-DP, here

$$\sigma = c_2 \frac{q\sqrt{T_i \log(1/\delta)}}{\varepsilon} \tag{4}$$

It is noted that the sampling rate $q = 1$ if the edge server performs local training on all samples in each iteration, The details of Gaussian noise-based DP mechanism can be seen in [18].

Corollary 1 (Variance of Gaussian Noise):

$$\mathbb{E}\left\{\left\|\mathbf{n}^t\right\|^2\right\} = c_2^2 q^2 \log(1/\delta) \varphi^2 \frac{NTD}{d^2 \varepsilon^2 S} \tag{5}$$

Here, $D = \sum_{i \in \mathcal{P}_t} d_i^2$. The specific proof is presented in Appendix D.

3) Convergence Analysis: According to the description of the algorithm, the global loss function can be defined as $\mathcal{L}(\cdot) = \sum_{i \in \mathcal{P}_t} p_i \mathcal{L}_i(\cdot)$, where $\mathcal{L}_i(\cdot)$ is the local loss function of the i-th edge server with a weight factor of $p_i = N d_i / S d$. The specific assumptions are firstly given as follows, so that it can provide the theoretical basis for the specific analysis.

Assumption 1: The local loss function $\mathcal{L}_i(\cdot)$ is convex.

Assumption 2: The loss function $\mathcal{L}_i(\cdot)$ is η-Lipschitz smooth. For any model parameters $\mathbf{w}^a$ and $\mathbf{w}^b$, there exists a constant $\eta > 0$, such that the following two inequalities hold.

$$\mathcal{L}_i(\mathbf{w}^a) \leq \mathcal{L}_i(\mathbf{w}^b) + \left\langle \nabla\mathcal{L}_i(\mathbf{w}^b), \mathbf{w}^a - \mathbf{w}^b \right\rangle + \frac{\eta}{2}\left\|\mathbf{w}^a - \mathbf{w}^b\right\|^2$$

$$\left\|\nabla\mathcal{L}_i(\mathbf{w}^a) - \nabla\mathcal{L}_i(\mathbf{w}^b)\right\| \leq \eta \cdot \left\|\mathbf{w}^a - \mathbf{w}^b\right\|$$

Assumption 3: There exists a constant $\mu > 0$ such that the Polyak-Lojasiewicz inequality $\mathcal{L}(\mathbf{w}) - \mathcal{L}(\mathbf{w}^*) \leq \frac{1}{2\mu}\|\nabla\mathcal{L}(\mathbf{w})\|^2$ holds when the model obtains the optimal solution $\mathbf{w}^*$.

Assumption 4: For any edge server i and model parameter $\mathbf{w}$, there exists a constant β_i, such that the deviation between the local gradient and the global gradient is bounded by β_i, i.e., $\|\nabla\mathcal{L}_i(\mathbf{w}) - \nabla\mathcal{L}(\mathbf{w})\| \leq \beta_i$.

Assumption 5: For any model parameter $\mathbf{w}$, there exists a constant λ such that the gradient is bounded by λ, i.e., $\|\nabla\mathcal{L}(\mathbf{w})\| \leq \lambda$.

By using these assumptions, the following lemmas and theorem are proposed, and then proved in the appendix.

Lemma 2 (κ-divergence of edge servers): If the model parameter $\mathbf{w}$ is known, there exist a constant κ such that

$$\mathrm{E}\left\{\|\nabla\mathcal{L}_i(\mathbf{w})\|^2\right\} \leq \|\nabla\mathcal{L}(\mathbf{w})\|^2 \kappa^2 \tag{6}$$

Inherently, the value κ is closely related to the loss function and the distribution of data samples. Lemma 2 is mainly based on the statistical heterogeneity of data in each edge server, which may lead to large differences in different local models. Therefore, loss functions of different local models are analyzed in order to find out the intrinsic correlation with the global loss function. The specific proof can be seen in Appendix E.

Lemma 3 (Expected deviation of two adjacent global loss functions): Given the Assumption 2, the expected deviation of the global loss functions in two adjacent rounds is defined as follows

$$\begin{aligned}\mathrm{E}\left\{\nabla\mathcal{L}(\hat{\mathbf{w}}^{t+1}) - \nabla\mathcal{L}(\hat{\mathbf{w}}^{t})\right\} &\leq \alpha_2\mathrm{E}\left\{\left\|\nabla\mathcal{L}(\hat{\mathbf{w}}^{t})\right\|^2\right\} \\ &+ \alpha_1\mathrm{E}\left\{\left\|\nabla\mathcal{L}(\hat{\mathbf{w}}^{t})\right\|\left\|\mathbf{n}^{t+1}\right\|\right\} + \alpha_0\mathrm{E}\left\{\left\|\mathbf{n}^{t+1}\right\|^2\right\}\end{aligned} \tag{7}$$

where the coefficients on the right-hand side of the inequality are respectively $\alpha_2 = \frac{\eta\kappa^2}{2\rho^2} + \frac{\eta\kappa}{\rho^2} - \frac{1}{\rho}$, $\alpha_1 = \frac{\eta\kappa}{\rho} + 1$ and $\alpha_0 = \frac{\eta}{2}$. $\mathbf{n}^{t+1}$ is the superposition of all noises generated by edge servers participating in the $(t+1)$-th global aggregation, and $\mathbf{n}^{t+1} = \sum_{i\in\mathcal{P}_t} p_i\mathbf{n}_i^{t+1}$, according to Algorithm 1, the weight coefficient is $p_i = Nd_i/Sd$. The specific proof can be seen in Appendix F.

Theorem 3 (The convergence upper bound of the algorithm): After T rounds of global aggregation, the convergence upper bound of the algorithm is as follows

$$\begin{aligned}\mathrm{E}\left\{\mathcal{L}(\hat{\mathbf{w}}^{\mathrm{T}}) - \mathcal{L}(\mathbf{w}^*)\right\} &\leq \mathrm{Z}^{\mathrm{T}}\Gamma \\ &+ \left\{\alpha_1\lambda\frac{H\sqrt{NT}}{\varepsilon\sqrt{S}} + \alpha_0\frac{H^2NT}{\varepsilon^2 S}\right\}\frac{Z^T - 1}{Z - 1}\end{aligned} \tag{8}$$

where the parameters Z, Γ and H are given by $Z = 1 + 2\mu\alpha_2$, $\Gamma \triangleq \mathrm{E}\{\mathcal{L}(\hat{\mathbf{w}}^0) - \mathcal{L}(\mathbf{w}^*)\}$ and $H = \frac{c_2 q\varphi\sqrt{D\log(1/\delta)}}{d}$.

The specific proof can be seen in Appendix G.

IV. Experiment and Result Analysis

In order to evaluate the proposed two-stage privacy-preserving FL framework, we adopt the ECG signals as dataset and start server/client (cloud server/edge server) programs on the Windows server for performance testing. For the ECG signal, we extract the instantaneous frequency (instfreq) and spectral entropy (pentropy) as features, which are then perturbed and trained by LSTM network. After training, the local model is superimposed by noise before being sent to the server. The proposed FL scheme is compared with the noise-free scheme so as to investigate the impact of perturbance and noise on the model convergence and accuracy, which can be reflected by adjusting different values of DP-related parameters and different combinations of two stages.

A. Dataset and Experiment Setting

ECG dataset employed in our experiment comes from the PhysioNet 2017 Challenge[1] [28], [29]. An ECG signal records the electrical activity of a person's heart during a period of time. According to ECG signal, whether heartbeat is normal or abnormal can be detected. For example, atrial fibrillation (AFib) is an abnormal heartbeat that occurs when atria lose coordination with ventricle. The training dataset consists of a set of ECG signals sampled at frequency 300 Hz, and they are annotated into two different types, namely normal (N) and AFib (A). For ECG classification, we firstly take time-frequency analysis and then use BiLSTM network to train a binary classifier that can distinguish normal ECG signals from AFib signals. We employ matlab for programming in 64-bit Windows 7 with Xeon E5-1650 V2 CPU and 32 GB DDR3 memory. We started 1 server program and 8 client programs to collaboratively train the classification model in parallel. The samples are independent and identically distributed (IID), and the data is randomly shuffled and distributed to 8 clients.

B. Feature Perturbation and Noise Addition

In this article, BiLSTM network is used to classify ECG signals. The LSTM network is a recurrent neural network (RNN) which is suitable for studying on sequential signal. The long-term correlations between time steps of sequences can be learned from LSTM network. LSTM layers can analyze time sequences forward, while BiLSTM layers can analyze time sequences forward and backward. In our experiment, the concrete procedures orderly include loading of original ECG signal, signal preprocessing, feature extraction, normalization,

[1][Online]. Available: https://physionet.org/challenge/2017/

perturbation, classifier training, addition of Gaussian noise and model aggregation.

1) Loading of ECG signals:

The original ECG dataset contains 738 AFib signals and 5050 normal signals. Most of the signals consist of 9000 sampling points in length.

2) Signal preprocessing:

Firstly, the ECG signal is divided into several segments for 9000 sampling points per segment. Next, the classifier is designed and the dataset is split into two sets, i.e., training set and testing set. Since the quantity ratio of the original AFib signal to normal signal is about 1:7, which illustrates the AFib samples is insufficient. Therefore, it is necessary to increase the number of AFib samples by using data augmentation. Finally, we get 48000 AFib samples and normal samples in the training set, and 4000 ones in the testing set.

3) Feature extraction:

Extracting effective features from signals is benefit for improving the accuracy of the classifier. Firstly, we calculate the spectrums of the ECG signal by using the short-time Fourier transform on a time window with window size 255, and then extract the time-frequency (TF) moment information from the spectrums. Each moment is in fact a one-dimension feature which can be used for input of the LSTM network. In this article, we extract two TF moments i.e., instfreq and pentropy, which are taken as samples for training dataset and testing dataset. According to feature extraction, a two-dimension sample no longer contains 9000 sampling points, but only 255 sampling points per dimension.

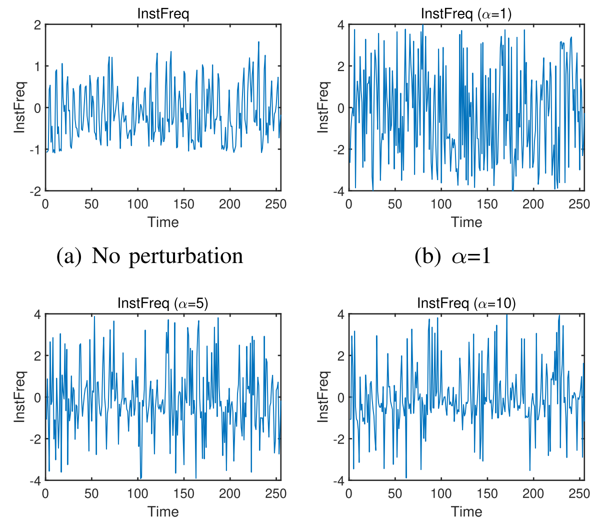
4) Feature normalization:

In instfreq or pentropy, there exists large deviations among different values, furthermore, the mean value of instfreq is probably too high. Thus, it inevitably causes inefficiency in model training by LSTM network. Thus, we normalize the training and testing datasets by using the mean value and standard deviation.

5) BiLSTM training network:

We design the model as five layers: sequence input layer with 2 dimensions, BiLSTM layer with 100 hidden units, 2 fully connected layer, 1 softmax layer and 1 classification output layer. Finally, the model outputs two classifications including normal ECG and AFib ECG. In the training process, we adopt SGD algorithm to train models and Cross-Entropy for classification, where, the training-related parameters are set as mini-batch size = 150, learning rate = 0.01. In order to avoid memory exhaustion due to process too much data at once, the signal is split into smaller pieces i.e., setting sequence length = 1000. Furthermore, we set gradient threshold = 1 so as to stabilize the training process by preventing the gradient from being too large. For detecting abnormal ECG signals effectively, oversampling is used to avoid classification bias.

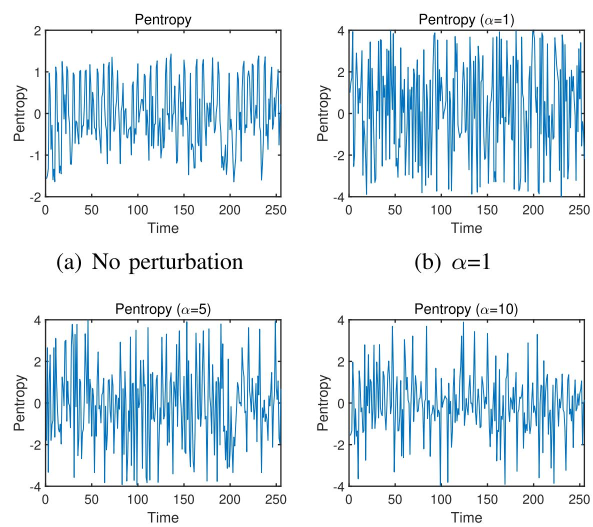
In the FL framework, the adjustable privacy-preserving parameters include in parameters ε_1 and α of the first stage, and parameters ε_2 and δ of the second stage. We give an instance of ECG instfreq and ECG pentropy to exhibit the effect of the features perturbation in the first stage, which is shown as Fig. 3 and Fig. 4. Here, the variation of effects can be obviously observed according to the three different values of perturbant coefficient α ($\alpha = 1$, $\alpha = 5$, $\alpha = 10$) and the case of no perturbation. From the comparison in the figure, it can be seen that the smaller the value of α, the larger the feature perturbation.



(a) No perturbation (b) α=1

(c) α=5 (d) α=10

Fig. 3. Comparison of instfreq perturbation with different parameter α.



(a) No perturbation (b) α=1

(c) α=5 (d) α=10

Fig. 4. Comparison of pentropy perturbation with different parameter α.

C. Evaluation and Discussion

1) The Impact of Parameter α on Accuracy: The effect of ECG signal feature perturbation on model accuracy is evaluated by only adjusting the values of parameters ε_1 and α involved in the first stage while not adding noise in the second stage. We set privacy budget ε_1 as a fixed value 10 and adjust the parameter α for different values to test the variation of model accuracy. As a comparison, the model accuracy of traditional FL with no any perturbance and noise is tested. The accuracy

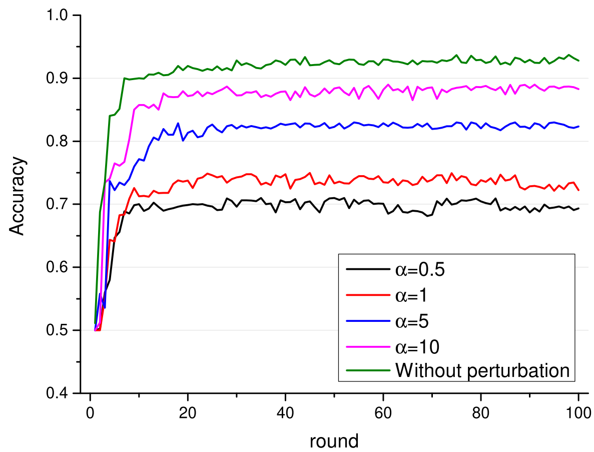


Fig. 5. Model accuracy with different parameter α.

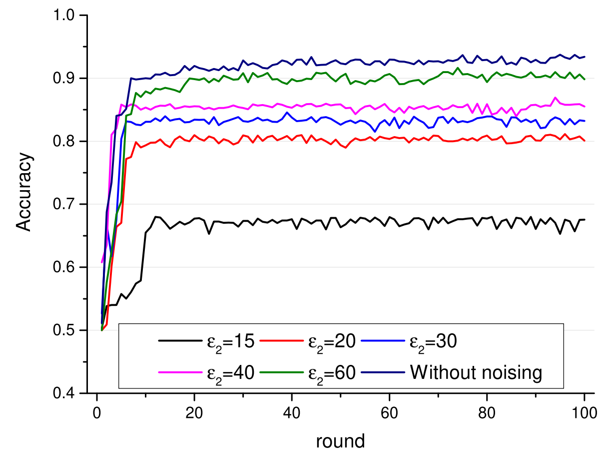


Fig. 6. Model accuracy with different parameter ε_2.

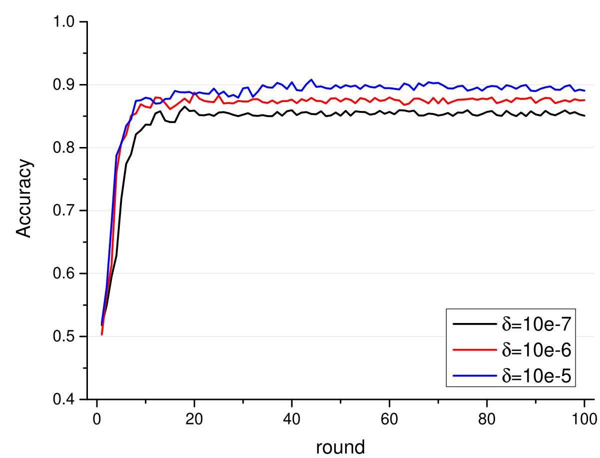


Fig. 7. Model accuracy with different parameter δ.

evaluation results of the model on the testing dataset are shown in Fig. 5, which exhibits the variations of accuracy in five conditions, (a) $\alpha = 0.5$, (b) $\alpha = 1$, (c) $\alpha = 5$, (d) $\alpha = 10$, and (e) without perturbation. It is clear from the comparison of different curves in Fig. 5 that the smaller the value of α, the larger the perturbation, which caused more impairment on the model accuracy. As the number of global rounds increases, the model accuracy gradually promotes and eventually maintains a certain level. In the theoretical analysis, we found that the sensitivity of the feature data is large, which makes the effect of ε_1 on the accuracy very small. Therefore, parameter ε_1 is set as a fixed value instead of being adjusted in the experiments. The autonomous control of the privacy preserving level can be realized by adjusting α for different terminal users.

2) The Impact of Parameter ε_2 and δ on Accuracy: With no feature perturbation in the first stage, Gaussian noise is added to the local model in the second stage. Firstly, for evaluating the effect of privacy budgets on model training, we fix the value of slackness as $\delta = 10e\text{-}6$ and set some relatively large ε_2 for the model due to its high dimensionality. We test five different values of parameter ε_2, i.e., $\varepsilon_2 = 15$, 20, 30, 40, and 60. For comparison, the case of no noise addition is also evaluated. The model accuracy evaluation results on the testing dataset are shown in Fig. 6. The results demonstrate that the model accuracy is significantly affected by the noise intensity, and the larger ε_2 is, the higher the model accuracy is. When $\varepsilon_2 = 15$, the variance of Gaussian noise is up to a large degree, and produces gradient explosion in the training process. For solving this problem, gradient clipping is employed, which leads to a poor training effect to some extent, and the maximum model accuracy can only reach 67%. Subsequently, we fix the value of privacy budget as $\varepsilon_2 = 50$ and change the value of slackness as $\delta = 10e\text{-}5$, 10e-6, 10e-7, then, the model accuracy evaluation results are shown in Fig. 7. It can be seen from the Fig. 7 that the smaller the value of parameter δ, the larger the noise variance and the lower the model accuracy, but the effect of slackness δ on accuracy is not as significant as that of privacy budget ε_2. When adding Gaussian noise in the second stage, ε_2 is smaller, the loss of accuracy is larger.

TABLE II
FOUR TYPICAL SETS WITH DIFFERENT PARAMETER VALUES

Scenario	ε_1	α	ε_2	δ
(a)	10	8	50	10e-6
(b)	10	8	60	10e-6
(c)	10	12	50	10e-5
(d)	10	12	60	10e-5

3) The Impact of the Combination of Feature Perturbance and Noise Addition on Accuracy: The effect on the model accuracy is evaluated by combining the first and second stages, i.e., by simultaneously perturbing the signal features and adding Gaussian noise to the local model. We selected four typical sets of parameters ε_1, α, ε_2, and δ which are shown in Table II. These four combined sets are compared with the case of no feature perturbation and no noise addition. As the number of aggregations increases, the variation of model accuracy is shown in Fig. 8. Here, the green curve represents the accuracy of the model trained in the case of no any privacy preserving measures, which is taken as reference benchmark. Comparing different curves, it can be found that the proposed combinative mechanism with perturbation and noise addition retains the

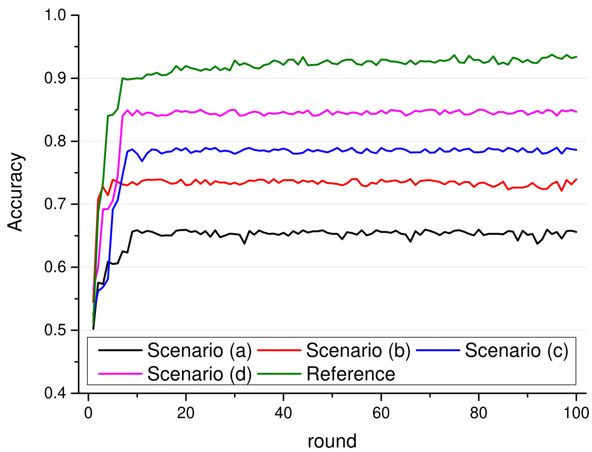


Fig. 8. Model accuracy in two stage combinations with different parameter values.

convergence characteristics of the general model, but accuracies of all classifiers are lower than the reference benchmark. The accuracy obtained in scenario (a) is the least effective, while the accuracy in scenario (d) is the best except the benchmark. There is a worse performance in combination scenario of two stages compared to the scenario of a single stage.

V. Conclusion

This article focuses on how to effectively combine DP technology and FL technology to promote data privacy security in smart healthcare scenarios. In the article, A two-stage DP-based FL framework orienting edge intelligence is proposed. In collaborative end-edge -cloud FL, the training samples come from terminal users who can collect physiological signals by the help of wearable devices. However, the training tasks are performed by edge servers and cloud servers instead of the terminal users. The framework can provide users with diverse privacy preserving levels depending on the sensitivity of data. In the first stage randomized response mechanism is adopted at terminal users for perturbing the feature data, which can facilitate adaptive adjustment of the privacy preservation level for users. In the second stage, noise is imposed on the models to further enhance the privacy security of models. To evaluate the performance of the proposed framework theoretically, model convergence is analyzed in detail. The experiment is executed on a real ECG signal dataset by employing a FL-based BiLSTM network model and a Gaussian noise addition mechanism. By setting different parameter values and adjusting the different combination of feature perturbation and noise addition, the variation of model accuracy is tested. The experimental results demonstrate that the proposed framework provides good accuracy and convergence while ensuring privacy.

Appendix A
Proof of DP

Proof: Considering a sensitivity of $\Delta f = cn$, choose p and q as follows,

$$p = \frac{e^{\varepsilon/cn}}{1+e^{\varepsilon/cn}}, q = \frac{1}{1+e^{\varepsilon/cn}},$$

$$\begin{aligned}\frac{\Pr[X|s_1]}{\Pr[X|s_2]} &= \frac{\prod_{i\in[cn]}\Pr[X[i]|s_1]}{\prod_{i\in[cn]}\Pr[X[i]|s_2]}\\ &\le \left(\frac{\Pr[X[s_1]=1|s_1]\Pr[X[s_2]=0|s_1]}{\Pr[X[s_1]=1|s_2]\Pr[X[s_2]=0|s_2]}\right)^{\frac{cn}{2}}\\ &= \left(\frac{p}{q}\cdot\frac{1-q}{1-p}\right)^{\frac{cn}{2}} = e^{\varepsilon} \end{aligned} \tag{9}$$

The proof is completed.

Appendix B
Proof of Theorem 1

Proof: Considering sensitivity $\Delta f = cn$, giving different perturbance probabilities for bit 1 and bit 0 as follows.

$$\Pr[X[s_1]=1|s_1] = \frac{1}{1+\alpha}, \Pr[X[s_1]=1|s_2] = \frac{1}{1+\alpha e^{\frac{2\varepsilon}{cn}}},$$

$$\Pr[X[s_2]=0|s_1] = \frac{\alpha e^{\frac{2\varepsilon}{cn}}}{1+\alpha e^{\frac{2\varepsilon}{cn}}}, \Pr[X[s_2]=0|s_2] = \frac{\alpha}{1+\alpha},$$

Therefore,

$$\begin{aligned}\frac{\Pr[X|s_1]}{\Pr[X|s_2]} &= \frac{\prod_{i\in[cn]}\Pr[X[i]|s_1]}{\prod_{i\in[cn]}\Pr[X[i]|s_2]}\\ &\le \left(\frac{\Pr[X[s_1]=1|s_1]\Pr[X[s_2]=0|s_1]}{\Pr[X[s_1]=1|s_2]\Pr[X[s_2]=0|s_2]}\right)^{\frac{cn}{2}}\\ &= \left(\frac{\left(\frac{1}{1+\alpha}\right)}{\left(\frac{1}{1+\alpha e^{\frac{2\varepsilon}{cn}}}\right)}\cdot\frac{\left(\frac{\alpha e^{\frac{2\varepsilon}{cn}}}{1+\alpha e^{\frac{2\varepsilon}{cn}}}\right)}{\left(\frac{\alpha}{1+\alpha}\right)}\right)^{\frac{cn}{2}} = e^{\varepsilon}\end{aligned} \tag{10}$$

The proof of Theorem 1 is completed.

Appendix C
Proof of Lemma 1

Proof. In Algorithm 1, the cloud server uses robin mechanism to select participants for each global aggregation, and S edge servers are selected in each round, where there are N edge servers in total. Thus, there exists C_N^S different selections, the set of all selections is denoted as $\phi_S(N)$. For instance, supposing the set of all edge servers is $\mathcal{P} = \{a, b, c\}$ and $S = 2$, then, it can be known that $\phi_S(N) = \{\{a, b\}, \{b, c\}, \{a, c\}\}$ and $C_N^S = 3$. The robin mechanism conforms to the fairness principle, that means all of edge servers are selected with the same probability, so the total rounds of aggregations each edge server participates is considered to be equal. Considering there are C_N^S selections and S edge servers in each selection, so, all selections corresponding to $C_N^S \cdot S$ edge servers, while the total number of edge servers is N, it can be concluded that each edge server is selected for $C_N^S \cdot S/N$ times in average. Then, the following equation can be deduced.

$$\sum_{\mathcal{P}_t\in\phi_S(N)}\sum_{i\in\mathcal{P}_t}\frac{d_i}{d}\mathbf{w}_i^t=\frac{C_N^S S}{N}\sum_{i\in\mathcal{P}}\frac{d_i}{d}\mathbf{w}_i^t \tag{11}$$

By Equation (11), it can be calculated:

$$\begin{aligned}\mathrm{E}\{\bar{\mathbf{w}}^{\mathrm{t}}_{\mathrm{i}\in\mathcal{P}_{\mathrm{t}}}\} &= \frac{N}{S}\mathrm{E}\left\{\sum_{i\in\mathcal{P}_t}\frac{d_i}{d}\mathbf{w}_i^t\right\}\\ &= \frac{N}{S}\sum_{\mathcal{P}_t\in\phi_S(N)}\frac{1}{C_N^S}\sum_{i\in\mathcal{P}_t}\frac{d_i}{d}\mathbf{w}_i^t\\ &= \sum_{i\in\mathcal{P}}\frac{d_i}{d}\mathbf{w}_i^t\\ &= E\{\bar{\mathbf{w}}^t_{i\in\mathcal{P}}\}\end{aligned} \tag{12}$$

The proof of Lemma 1 is completed.

Appendix D
Proof of Corollary 1

Proof: To obtain the total noise variance for each global aggregation in Algorithm 1, it is assumed that the noise of each edge server is IID in distribution. By using the Gaussian mechanism in Theorem 2, the aggregated noise variance is as equation (13).

$$\begin{aligned}\mathbb{E}\left\{\left\|\mathbf{n}^t\right\|^2\right\} &= \mathbb{E}\left\{\left\|\frac{N}{S}\sum_{i\in\mathcal{P}_t}\frac{d_i}{d}\mathbf{n}_i^t\right\|^2\right\}\\ &= \frac{N^2}{S^2\,d^2}\sum_{i\in\mathcal{P}_t}d_i^2\mathbb{E}\left\{\left\|\mathbf{n}_i^t\right\|^2\right\}\\ &= \frac{N^2}{S^2\,d^2}\cdot c^2q^2\log(1/\delta)\frac{\varphi^2T_i}{\varepsilon^2}\cdot\sum_{i\in\mathcal{P}_t}d_i^2\\ &= c_2^2q^2\log(1/\delta)\varphi^2\frac{NTD}{d^2\varepsilon^2 S}\end{aligned} \tag{13}$$

The proof of Corollary 1 is completed.

Appendix E
Proof of Lemma 2

Proof: Let's firstly calculate the variance between the local gradient and the global gradient. According to the definition of expectation, it can be known $\mathrm{E}\{\nabla\mathcal{L}_i(\mathbf{w})\}=\nabla\mathcal{L}(\mathbf{w})$. Then,

$$\begin{aligned}&\mathrm{E}\left\{\left\|\nabla\mathcal{L}_{\mathrm{i}}(\mathbf{w})-\nabla\mathcal{L}(\mathbf{w})\right\|^2\right\}\\ &=\mathrm{E}\left\{\left\|\nabla\mathcal{L}_i(\mathbf{w})\right\|^2\right\}-2\mathrm{E}\left\{\langle\nabla\mathcal{L}_i(\mathbf{w}),\nabla\mathcal{L}(\mathbf{w})\rangle\right\}+\left\|\nabla\mathcal{L}(\mathbf{w})\right\|^2\\ &=\mathrm{E}\left\{\left\|\nabla\mathcal{L}_i(\mathbf{w})\right\|^2\right\}-\left\|\nabla\mathcal{L}(\mathbf{w})\right\|^2\end{aligned} \tag{14}$$

By Assumption 4, $\mathrm{E}\{\|\nabla\mathcal{L}_i(\mathbf{w})-\nabla\mathcal{L}(\mathbf{w})\|^2\}\le\mathrm{E}\{\beta_i^2\}$, substituting into equation (14), gets

$$\mathrm{E}\left\{\left\|\nabla\mathcal{L}_i(\mathbf{w})\right\|^2\right\}\le\mathrm{E}\left\{\beta_i^2\right\}+\left\|\nabla\mathcal{L}(\mathbf{w})\right\|^2 \tag{15}$$

We let $\kappa(\mathbf{w})=\sqrt{\frac{\mathrm{E}\{\beta_i^2\}}{\|\nabla\mathcal{L}(\mathbf{w})\|^2}+1}$, the smaller the deviation of the local gradient from the global gradient, the smaller the $\kappa(w)$. Because loss function $\mathcal{L}(\mathbf{w})$ is correlated with the empirical risk objective, considering the extreme case, if the data of all edge servers are IID and sampled with the same probability, when all of loss functions converge to the same expected risk value, then $\mathrm{E}\{\beta_i^2\}=0$, $\kappa(\mathbf{w})$ reaches a minimum value 1. However, the data of different edge servers is usually heterogeneous. Even if the samples are homogeneous, it is difficult to reach a minimum value 1 for $\kappa(\mathbf{w})$ due to the sampling differences caused by finiteness of sample quantity. Therefore, it is necessary to define divergence whether it is IID-based FL or non-IID-based FL. Particularly, it is noted that the divergence $\kappa(\mathbf{w})$ is still not too large during the model training even though the data samples are heterogenous. Therefore, we assume that the upper bound of $\kappa(\mathbf{w})$ is κ, then

$$\mathrm{E}\left\{\left\|\nabla\mathcal{L}_i(\mathbf{w})\right\|^2\right\}\le\left\|\nabla\mathcal{L}(\mathbf{w})\right\|^2\kappa^2 \tag{16}$$

The proof of Lemma 2 is completed.

Appendix F
Proof of Lemma 3

Proof. According to Assumption 2, given any $\hat{\mathbf{w}}^{t+1}$ and $\hat{\mathbf{w}}^t$, taking the expectation of the first inequality defined by the η-Lipschitz smooth, then, the inequation (17) is obtained.

$$\begin{aligned}\mathrm{E}\left\{\mathcal{L}_{\mathrm{i}}(\hat{\mathbf{w}}^{\mathrm{t+1}})-\mathcal{L}_{\mathrm{i}}(\hat{\mathbf{w}}^{\mathrm{t}})\right\}&\le\mathrm{E}\{\langle\nabla\mathcal{L}_{\mathrm{i}}(\mathbf{w}^{\mathrm{t}}),\hat{\mathbf{w}}^{\mathrm{t+1}}-\hat{\mathbf{w}}^{\mathrm{t}}\rangle\}\\ &+\frac{\eta}{2}\mathrm{E}\left\{\left\|\hat{\mathbf{w}}^{t+1}-\hat{\mathbf{w}}^t\right\|^2\right\}\end{aligned} \tag{17}$$

By the definition of expectation, there exists $\mathrm{E}\{\mathcal{L}_i(\hat{\mathbf{w}}^t)\}=\mathcal{L}(\hat{\mathbf{w}}^t)$. Then, the inequation (17) can be converted as in equation (18).

$$\begin{aligned}\mathrm{E}\left\{\mathcal{L}(\hat{\mathbf{w}}^{\mathrm{t+1}})-\mathcal{L}(\hat{\mathbf{w}}^{\mathrm{t}})\right\}&\le\mathrm{E}\left\{\langle\nabla\mathcal{L}(\hat{\mathbf{w}}^{\mathrm{t}}),\hat{\mathbf{w}}^{\mathrm{t+1}}-\hat{\mathbf{w}}^{\mathrm{t}}\rangle\right\}\\ &+\frac{\eta}{2}\mathrm{E}\left\{\left\|\hat{\mathbf{w}}^{t+1}-\hat{\mathbf{w}}^t\right\|^2\right\}\end{aligned} \tag{18}$$

For simplification of description, the loss function with regularization term is defined as follow.

$$\psi(\mathbf{w};\hat{\mathbf{w}}^t)\triangleq\mathcal{L}_i(\mathbf{w})+\frac{\rho}{2}\left\|\mathbf{w}-\hat{\mathbf{w}}^t\right\|^2 \tag{19}$$

Taking derivative of the above definition, then we get

$$\nabla\psi(\mathbf{w};\hat{\mathbf{w}}^t)=\nabla\mathcal{L}_i(\mathbf{w})+\rho\left\|\mathbf{w}-\hat{\mathbf{w}}^t\right\| \tag{20}$$

Referring to the definition given in [30], there exists a positive number $\tau\in[0,1]$, when $\mathbf{w}_i^{t+1}$ is a τ-inexact solution of the function $\min_{\mathbf{w}}\psi(\mathbf{w},\hat{\mathbf{w}}^t)$, the following inequation holds.

$$\begin{aligned}\mathrm{E}\left\{\left\|\nabla\psi(\mathbf{w}_{\mathrm{i}}^{\mathrm{t+1}};\hat{\mathbf{w}}^{\mathrm{t}})\right\|\right\}&\le\tau\mathrm{E}\left\{\left\|\nabla\psi(\hat{\mathbf{w}}^{\mathrm{t}},\hat{\mathbf{w}}^{\mathrm{t}})\right\|\right\}\\ &=\tau\mathrm{E}\left\{\left\|\nabla\mathcal{L}_i(\hat{\mathbf{w}}^t)\right\|\right\}\\ &\le\tau\kappa\left\|\nabla\mathcal{L}(\hat{\mathbf{w}}^t)\right\|\end{aligned} \tag{21}$$

Here, Lemma 2 is used in inequation (21). According to [30], supposing $\tilde{\mathbf{w}}_i^{t+1}=\arg\min_{\mathbf{w}}\psi_i(\mathbf{w},\hat{\mathbf{w}}^t)$, there exists a constant

$\bar{\rho} = \rho + \varsigma > 0$, such that $\psi(\mathbf{w}, \hat{\mathbf{w}}^t)$ is $\bar{\rho}$-convex. Then, the following two inequations hold.

$$\left\|\tilde{\mathbf{w}}_i^{t+1} - \mathbf{w}_i^{t+1}\right\| \le \frac{\tau}{\bar{\rho}} \left\|\nabla \mathcal{L}_i(\hat{\mathbf{w}}^t)\right\|$$
$$\left\|\tilde{\mathbf{w}}_i^{t+1} - \hat{\mathbf{w}}^t\right\| \le \frac{1}{\bar{\rho}} \left\|\nabla \mathcal{L}_i(\hat{\mathbf{w}}^t)\right\| \tag{22}$$

By using the above inequations and Lemma 2, we can deduce that

$$\begin{aligned} \mathrm{E}\left\{\left\|\mathbf{w}_{\mathrm{i}}^{t+1} - \hat{\mathbf{w}}^{\mathrm{t}}\right\|\right\} &= \mathrm{E}\left\{\left\|(\tilde{\mathbf{w}}_{\mathrm{i}}^{t+1} - \hat{\mathbf{w}}^t) - (\tilde{\mathbf{w}}_{\mathrm{i}}^{t+1} - \mathbf{w}_{\mathrm{i}}^{t+1})\right\|\right\} \\ &\le \mathrm{E}\left\{\left\|\tilde{\mathbf{w}}_i^{t+1} - \hat{\mathbf{w}}^t\right\| + \left\|\tilde{\mathbf{w}}_i^{t+1} - \mathbf{w}_i^{t+1}\right\|\right\} \\ &\le \frac{1+\tau}{\bar{\rho}} \mathrm{E}\left\{\left\|\nabla \mathcal{L}_i(\hat{\mathbf{w}}^t)\right\|\right\} \\ &\le \frac{(1+\tau)\kappa}{\bar{\rho}} \left\|\nabla \mathcal{L}(\hat{\mathbf{w}}^t)\right\| \end{aligned} \tag{23}$$

According Algorithm 1, we calculate the deviation between the model parameters of two adjacent rounds with superimposed noise. By using $\mathbf{w}^{t+1} = \sum_{i \in \mathcal{P}_t} p_i \mathbf{w}_i^{t+1} = \mathrm{E}(\mathbf{w}_i^{t+1})$, here, $p_i = N d_i / S d$, we can obtain

$$\begin{aligned} \left\|\hat{\mathbf{w}}^{t+1} - \hat{\mathbf{w}}^t\right\| &= \left\|\mathbf{w}^{t+1} + \mathbf{n}^{t+1} - \hat{\mathbf{w}}^t\right\| \\ &\le \mathrm{E}\left\{\left\|\mathbf{w}_i^{t+1} - \hat{\mathbf{w}}^t\right\|\right\} + \mathrm{E}\left\{\left\|\mathbf{n}^{t+1}\right\|\right\} \\ &\le \frac{(1+\tau)\kappa}{\bar{\rho}} \left\|\nabla \mathcal{L}(\hat{\mathbf{w}}^t)\right\| + \left\|\mathbf{n}^{t+1}\right\| \end{aligned} \tag{24}$$

Now, we define Φ^{t+1} as follow.

$$\Phi^{t+1} \triangleq \nabla \mathcal{L}(\hat{\mathbf{w}}^t) + \rho\left(\mathbf{w}^{t+1} - \hat{\mathbf{w}}^t\right) \tag{25}$$

According to Assumption 2, as well combining with inequation (21) and (23), we can extrapolate $\|\Phi^{t+1}\|$ as follow.

$$\begin{aligned} \left\|\Phi^{t+1}\right\| &= \left\|\nabla \mathcal{L}(\hat{\mathbf{w}}^t) + \rho(\mathbf{w}^{t+1} - \hat{\mathbf{w}}^t)\right\| \\ &= \mathrm{E}\left\{\left\|\nabla \mathcal{L}_i(\hat{\mathbf{w}}^t) + \rho(\mathbf{w}_i^{t+1} - \hat{\mathbf{w}}^t)\right\|\right\} \\ &= \mathrm{E}\left\{\left\|\nabla \mathcal{L}_i(\mathbf{w}_i^{t+1}) - \nabla \mathcal{L}_i(\hat{\mathbf{w}}^t) - \left(\nabla \mathcal{L}_i(\mathbf{w}_i^{t+1})\right.\right.\right. \\ &\quad \left.\left.\left. + \ \rho(\mathbf{w}_i^{t+1} - \hat{\mathbf{w}}^t)\right)\right\|\right\} \\ &= \mathrm{E}\left\{\left\|\nabla \mathcal{L}_i(\mathbf{w}_i^{t+1}) - \nabla \mathcal{L}_i(\hat{\mathbf{w}}^t) - \nabla \psi(\mathbf{w}_i^{t+1}; \hat{\mathbf{w}}^t)\right\|\right\} \\ &\le \eta \mathrm{E}\left\{\left\|\mathbf{w}_i^{t+1} - \hat{\mathbf{w}}^t\right\|\right\} + \mathrm{E}\left\{\left\|\nabla \psi(\mathbf{w}_i^{t+1}; \hat{\mathbf{w}}^t)\right\|\right\} \\ &\le \frac{\eta(1+\tau)\kappa}{\bar{\rho}} \left\|\nabla \mathcal{L}(\hat{\mathbf{w}}^t)\right\| + \tau\kappa \left\|\nabla \mathcal{L}(\hat{\mathbf{w}}^t)\right\| \end{aligned} \tag{26}$$

By using inequation (26), and considering the addition of the noise term to equation (25), inequation (27) can be calculated as follow.

$$\begin{aligned} &\left\|\nabla \mathcal{L}(\hat{\mathbf{w}}^t) + \rho(\hat{\mathbf{w}}^{t+1} - \hat{\mathbf{w}}^t)\right\| \\ &= \left\|\nabla \mathcal{L}(\hat{\mathbf{w}}^t) + \rho(\mathbf{w}^{t+1} - \hat{\mathbf{w}}^t) + \rho \mathbf{n}^{t+1}\right\| \\ &= \left\|\Phi^{t+1} + \rho \mathbf{n}^{t+1}\right\| \\ &\le \left\|\Phi^{t+1}\right\| + \rho \left\|\mathbf{n}^{t+1}\right\| \end{aligned} \tag{27}$$

Substituting inequation (27) and (24) into inequation (18), then, we can deduce

$$\begin{aligned} &\mathrm{E}\left\{\mathcal{L}(\hat{\mathbf{w}}^{t+1}) - \mathcal{L}(\hat{\mathbf{w}}^t)\right\} \\ &\le \mathrm{E}\left\{\left\langle \nabla \mathcal{L}(\hat{\mathbf{w}}^t), \hat{\mathbf{w}}^{t+1} - \hat{\mathbf{w}}^t \right\rangle\right\} + \frac{\eta}{2} \mathrm{E}\left\{\left\|\hat{\mathbf{w}}^{t+1} - \hat{\mathbf{w}}^t\right\|^2\right\} \\ &\le \mathrm{E}\left\{\left\langle \nabla \mathcal{L}(\hat{\mathbf{w}}^t), \frac{1}{\rho}\left\|\Phi^{t+1}\right\| + \left\|\mathbf{n}^{t+1}\right\| - \frac{1}{\rho}\left\|\nabla \mathcal{L}(\hat{\mathbf{w}}^t)\right\| \right\rangle\right\} \\ &\quad + \frac{\eta}{2} \mathrm{E}\left\{\left(\frac{(1+\tau)\kappa}{\bar{\rho}} \left\|\nabla \mathcal{L}(\hat{\mathbf{w}}^t)\right\| + \left\|\mathbf{n}^{t+1}\right\|\right)^2\right\} \\ &\le \mathrm{E}\left\{\left\langle \nabla \mathcal{L}(\hat{\mathbf{w}}^t), \frac{\eta(1+\tau)\kappa}{\rho\bar{\rho}} \left\|\nabla \mathcal{L}(\hat{\mathbf{w}}^t)\right\| + \frac{\tau\kappa}{\rho}\left\|\nabla \mathcal{L}(\hat{\mathbf{w}}^t)\right\| \right.\right. \\ &\quad \left.\left. + \left\|\mathbf{n}^{t+1}\right\| - \frac{1}{\rho}\left\|\nabla \mathcal{L}(\hat{\mathbf{w}}^t)\right\| \right\rangle\right\} \\ &\quad + \frac{\eta}{2} \mathrm{E}\left\{\left(\frac{(1+\tau)\kappa}{\bar{\rho}} \left\|\nabla \mathcal{L}(\hat{\mathbf{w}}^t)\right\| + \left\|\mathbf{n}^{t+1}\right\|\right)^2\right\} \\ &= \alpha_2 \mathrm{E}\left\{\left\|\nabla \mathcal{L}(\hat{\mathbf{w}}^t)\right\|^2\right\} + \alpha_1 \mathrm{E}\left\{\left\|\nabla \mathcal{L}(\hat{\mathbf{w}}^t)\right\| \left\|\mathbf{n}^{t+1}\right\|\right\} \\ &\quad + \alpha_0 \mathrm{E}\left\{\left\|\mathbf{n}^{t+1}\right\|^2\right\} \end{aligned} \tag{28}$$

where the coefficients are $\alpha_2 = \frac{\eta(1+\tau)^2\kappa^2}{2\bar{\rho}^2} + \frac{\eta(1+\tau)\kappa}{\rho\bar{\rho}} + \frac{\tau\kappa}{\rho} - \frac{1}{\rho}$, $\alpha_1 = \frac{\eta(1+\tau)\kappa}{\bar{\rho}} + 1$, $\alpha_0 = \frac{\eta}{2}$, respectively. According to [30], when $\tau = 0$ and $\bar{\rho} = \rho$, all subproblems are solved for convex function $\psi(\mathbf{w}; \hat{\mathbf{w}}^t)$. At this point, $\alpha_2 = \frac{\eta\kappa^2}{2\rho^2} + \frac{\eta\kappa}{\rho^2} - \frac{1}{\rho}$, $\alpha_1 = \frac{\eta\kappa}{\rho} + 1$, $\alpha_0 = \frac{\eta}{2}$.

The proof of Lemma 3 is completed.

Appendix G
Proof of Theorem 3

Proof: Firstly, we take a slight transform of inequation (28) in Lemma 3, then obtain

$$\begin{aligned} &\mathrm{E}\left\{\mathcal{L}(\hat{\mathbf{w}}^{t+1}) - \mathcal{L}(\mathbf{w}^*) - \left(\mathcal{L}(\hat{\mathbf{w}}^t) - \mathcal{L}(\mathbf{w}^*)\right)\right\} \\ &\quad = \mathrm{E}\left\{\mathcal{L}(\hat{\mathbf{w}}^{t+1}) - \mathcal{L}(\hat{\mathbf{w}}^t)\right\} \\ &\quad \le \alpha_2 \mathrm{E}\left\{\left\|\nabla \mathcal{L}(\hat{\mathbf{w}}^t)\right\|^2\right\} + \alpha_1 \mathrm{E}\left\{\left\|\nabla \mathcal{L}(\hat{\mathbf{w}}^t)\right\| \left\|\mathbf{n}^{t+1}\right\|\right\} \\ &\quad\quad + \alpha_0 \mathrm{E}\left\{\left\|\mathbf{n}^{t+1}\right\|^2\right\} \end{aligned} \tag{29}$$

Combining with Assumption 3 and Assumption 5, inequation (30) can be furtherly deduced from inequation (29).

$$\begin{aligned} &\mathrm{E}\left\{\mathcal{L}(\hat{\mathbf{w}}^{\mathrm{t+1}}) - \mathcal{L}(\mathbf{w}^*)\right\} \\ &\quad \le \mathrm{E}\left\{\mathcal{L}(\hat{\mathbf{w}}^t) - \mathcal{L}(\mathbf{w}^*)\right\} + \alpha_2 \mathrm{E}\left\{\left\|\nabla \mathcal{L}(\hat{\mathbf{w}}^t)\right\|^2\right\} \\ &\quad\quad + \alpha_1 \mathrm{E}\left\{\left\|\nabla \mathcal{L}(\hat{\mathbf{w}}^t)\right\| \left\|\mathbf{n}^{t+1}\right\|\right\} + \alpha_0 \mathrm{E}\left\{\left\|\mathbf{n}^{t+1}\right\|^2\right\} \\ &\quad \le (1 + 2\mu\alpha_2)\, \mathrm{E}\left\{\mathcal{L}(\hat{\mathbf{w}}^t) - \mathcal{L}(\mathbf{w}^*)\right\} \end{aligned}$$

$$+\alpha_1\lambda \mathrm{E}\left\{\left\|\mathbf{n}^{t+1}\right\|\right\}+\alpha_0 \mathrm{E}\left\{\left\|\mathbf{n}^{t+1}\right\|^2\right\} \tag{30}$$

It is noted that inequation (30) is a recursive form on the $\mathrm{E}\{\mathcal{L}(\hat{\mathbf{w}}^{t+1})-\mathcal{L}(\mathbf{w}^*)\}$. Thus, after T rounds of global aggregation, we can get

$$\begin{aligned}
&\mathrm{E}\left\{\mathcal{L}(\hat{\mathbf{w}}^{\mathrm{T}})-\mathcal{L}(\mathbf{w}^*)\right\} \\
&\quad\leq (1+2\mu\alpha_2)^T \,\mathrm{E}\left\{\mathcal{L}(\hat{\mathbf{w}}^0)-\mathcal{L}(\mathbf{w}^*)\right\} \\
&\qquad+\left\{\alpha_1\lambda \mathrm{E}\left\{\left\|\mathbf{n}^{t+1}\right\|\right\}+\alpha_0 \mathrm{E}\left(\left\|\mathbf{n}^{t+1}\right\|^2\right)\right\}\sum_{0}^{T-1}(1+2\mu\alpha_2)^t \\
&\quad= (1+2\mu\alpha_2)^T \,\mathrm{E}\left\{\mathcal{L}(\hat{\mathbf{w}}^0)-\mathcal{L}(\mathbf{w}^*)\right\} \\
&\qquad+\left\{\alpha_1\lambda \mathrm{E}\left\{\left\|\mathbf{n}^{t+1}\right\|\right\}+\alpha_0 \mathrm{E}\left(\left\|\mathbf{n}^{t+1}\right\|^2\right)\right\} \\
&\quad\times \frac{(1+2\mu\alpha_2)^T-1}{2\mu\alpha_2}
\end{aligned} \tag{31}$$

Substituting the equation (5) in Corollary 1 into inequation (31), we can obtain conclusion as follow.

$$\begin{aligned}
&\mathrm{E}\left\{\mathcal{L}(\hat{\mathbf{w}}^{\mathrm{T}})-\mathcal{L}(\mathbf{w}^*)\right\} \leq \mathrm{Z}^{\mathrm{T}}\Gamma \\
&\quad+\left\{\alpha_1\lambda\frac{H\sqrt{NT}}{\varepsilon\sqrt{S}}+\alpha_0\frac{H^2NT}{\varepsilon^2 S}\right\}\frac{Z^T-1}{Z-1}
\end{aligned} \tag{32}$$

Here, we have $Z= 1+2\mu\alpha_2$, $\Gamma \triangleq \mathrm{E}\{\mathcal{L}(\hat{\mathbf{w}}^0)-\mathcal{L}(\mathbf{w}^*)\}$, $H=\frac{c_2 q\varphi\sqrt{D\log(1/\delta)}}{d}$.

The proof of Theorem 3 is completed.

References

[1] M. Jannesari et al., “Breast cancer histopathological image classification: A deep learning approach,” in *Proc. IEEE Int. Conf. Bioinf. Biomed.*, 2018, pp. 2405–2412.

[2] F. Murat, Ö. Yildirim, M. Talo, U. B. Baloglu, Y. Demir, and U. R. Acharya, “Application of deep learning techniques for heartbeats detection using ECG signals-analysis and review,” *Comput. Biol. Med.*, vol. 120, 2020, Art. no. 103726.

[3] B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. Aguera y Arcas, “Communication-efficient learning of deep networks from decentralized data,” in *Proc. 20th Int. Conf. Artif. Intell. Statist.*, 2017, pp. 1273–1282.

[4] Q. Yang, Y. Liu, T. Chen, and Y. Tong, “Federated machine learning: Concept and applications,” *ACM Trans. Intell. Syst. Technol.*, vol. 10, no. 2, pp. 12:1–12:19, 2019.

[5] J. Li et al., “A federated learning based privacy-preserving smart healthcare system,” *IEEE Trans. Ind. Inform.*, vol. 18, no. 3, pp. 2021–2031, Mar. 2022.

[6] Q. Wu, X. Chen, Z. Zhou, and J. Zhang, “Fedhome: Cloud-edge based personalized federated learning for in-home health monitoring,” *IEEE Trans. Mob. Comput.*, vol. 21, no. 8, pp. 2818–2832, Aug. 2022.

[7] J. Lee, J. Sun, F. Wang, S. Wang, C.-H. Jun, and X. Jiang, “Privacy-preserving patient similarity learning in a federated environment: Development and analysis,” *JMIR Med. Inform.*, vol. 6, no. 2, p. e20, 2018.

[8] L. Yuan, S. Zhang, G. Zhu, and K. Alinani, “Privacy-preserving mechanism for mixed data clustering with local differential privacy,” *Concurrency Computation: Pract. Experience*, 2022, doi: 10.1002/cpe.6503.

[9] X. Lyu et al., “Poisoning with cerberus: Stealthy and colluded backdoor attack against federated learning,” in *Proc. 37th AAAI Conf. Artif. Intell.*, 2023, pp. 9020–9028.

[10] X. Xu et al., “CGIR: Conditional generative instance reconstruction attacks against federated learning,” *IEEE Trans. Dependable Secure Comput.*, early access, Dec. 12, 2022, doi: 10.1109/TDSC.2022.3228302.

[11] P. Liu, X. Xu, and W. Wang, “Threats, attacks and defenses to federated learning: Issues, taxonomy and perspectives,” *Cybersecurity*, vol. 5, no. 1, pp. 1–19, 2022.

[12] W. Shi and S. Dustdar, “The promise of edge computing,” *Computer*, vol. 49, no. 5, pp. 78–81, 2016.

[13] Z. Zhou, X. Chen, E. Li, L. Zeng, K. Luo, and J. Zhang, “Edge intelligence: Paving the last mile of artificial intelligence with edge computing,” *Proc. IEEE*, vol. 107, no. 8, pp. 1738–1762, Aug. 2019.

[14] A. Appathurai et al., “A study on ECG signal characterization and practical implementation of some ECG characterization techniques,” *Measurement*, vol. 147, 2019, Art. no. 106384.

[15] P. Huang, L. Guo, M. Li, and Y. Fang, “Practical privacy-preserving ECG-based authentication for IoT-based healthcare,” *IEEE Internet Things J.*, vol. 6, no. 5, pp. 9200–9210, Oct. 2019.

[16] C. Dwork and A. Roth, “The algorithmic foundations of differential privacy,” *Found. Trends Theor. Comput. Sci.*, vol. 9, no. 3/4, pp. 211–407, 2014.

[17] H. Zheng, H. Hu, and Z. Han, “Preserving user privacy for machine learning: Local differential privacy or federated machine learning?,” *IEEE Intell. Syst.*, vol. 35, no. 4, pp. 5–14, 2020.

[18] M. Abadi et al., “Deep learning with differential privacy,” in *Proc. ACM SIGSAC Conf. Comput. Commun. Secur.*, 2016, pp. 308–318.

[19] Ú. Erlingsson, V. Pihur, and A. Korolova, “RAPPOR: Randomized aggregatable privacy-preserving ordinal response,” in *Proc. ACM SIGSAC Conf. Comput. Commun. Secur.*, 2014, pp. 1054–1067.

[20] W. Huang, S. Zhou, Y. Liao, and H. Chen, “An efficient differential privacy logistic classification mechanism,” *IEEE Internet Things J.*, vol. 6, no. 6, pp. 10620–10626, Dec. 2019.

[21] C. Xia, J. Hua, W. Tong, and S. Zhong, “Distributed K-means clustering guaranteeing local differential privacy,” *Comput. Secur.*, vol. 90, 2020, Art. no. 101699.

[22] H. Shin, S. Kim, J. Shin, and X. Xiao, “Privacy enhanced matrix factorization for recommendation with local differential privacy,” *IEEE Trans. Knowl. Data Eng.*, vol. 30, no. 9, pp. 1770–1782, Sep. 2018.

[23] R. C. Geyer, T. Klein, and M. Nabi, “Differentially private federated learning: A client level perspective,” 2017, *arXiv:1712.07557*.

[24] R. Hu, Y. Guo, H. Li, Q. Pei, and Y. Gong, “Personalized federated learning with differential privacy,” *IEEE Internet Things J.*, vol. 7, no. 10, pp. 9530–9539, Oct. 2020.

[25] L. Sun, J. Qian, and X. Chen, “LDP-FL: Practical private aggregation in federated learning with local differential privacy,” in *Proc. 30th Int. Joint Conf. Artif. Intell.*, 2021, pp. 1571–1578.

[26] K. Wei et al., “Federated learning with differential privacy: Algorithms and performance analysis,” *IEEE Trans. Inf. Forensics Secur.*, vol. 15, pp. 3454–3469, 2020.

[27] T. Wang, J. Blocki, N. Li, and S. Jha, “Locally differentially private protocols for frequency estimation,” in *Proc. 26th USENIX Secur. Symp.*, 2017, pp. 729–745.

[28] G. D. Clifford et al., “AF classification from a short single lead ECG recording: The Physionet computing in cardiology challenge 2017,” in *Proc. Comput. Cardiol.*, 2017, pp. 1–4.

[29] A. L. Goldberger et al., “Physiobank, Physiotoolkit, and Physionet: Components of a new research resource for complex physiologic signals,” *Circulation*, vol. 101, no. 23, pp. e215–e220, 2000.

[30] A. K. Sahu, T. Li, M. Sanjabi, M. Zaheer, A. Talwalkar, and V. Smith, “On the convergence of federated optimization in heterogeneous networks,” 2018, *arXiv:1812.06127*.